

Qeuph QUH

A Quantum-Resistant Blockchain Protocol

Qeuph secures every transaction with ML-DSA-87 lattice signatures (FIPS 204) and double SHA3-512 hashing, issues its 31.5 million QUH through the two-thirding reward schedule, and encodes addresses in Bech32m — a ledger designed to stay verifiable in the quantum era.

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Hybrid port of the QRL node architecture

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Abstract

As quantum computing advances, it poses a significant threat to traditional cryptographic systems, including those used in blockchain technology. Qeuph QUH is a pioneering blockchain protocol designed to address these vulnerabilities. By integrating the FIPS 204 Module-Lattice-Based Digital Signature Standard (ML-DSA-87), SHA3-512 hashing, and Bech32m addresses, Qeuph QUH offers unparalleled security, efficiency, and scalability. Compared to other quantum-resistant protocols, Qeuph QUH uses proven mechanisms such as “two-thirding” for block rewards, robust transaction validation, and quantum-resistant address generation. This whitepaper details the technical underpinnings and unique advantages of Qeuph QUH, setting a new standard for secure decentralized systems in the quantum era.

The design described here is fully implemented in an open, self-contained node suite: a pure-Python FIPS 204 reference implementation (cross-verified against OpenSSL’s ML-DSA), a proof-of-work daemon with peer-to-peer sync, a solo miner, an encrypted seed wallet, and a JSON-RPC interface. Where this paper states a rule, the reference implementation enforces it; Appendix A pins the exact parameters shipped in the code.

Executive Summary

Qeuph QUH is a quantum-resistant blockchain protocol addressing the vulnerabilities of current systems against quantum attacks. Its four pillars each solve a concrete failure mode of pre-quantum chains, and together they form a ledger that is intended to remain sound long after large-scale quantum computers arrive.

- **Quantum resistance.** Every signature on the chain is an ML-DSA-87 (FIPS 204, Category 5) lattice signature, and every identifier - transaction ids, block ids, addresses, and the proof-of-work digest - is a double SHA3-512 hash. No elliptic-curve operation and no SHA-2 commitment appears anywhere in the consensus path, so Shor- and Grover-class speedups do not weaken the security model.
- **Efficient transactions.** The UTXO ledger is enhanced with the txnonce field, a per-address strictly-increasing counter that makes transaction ordering deterministic and replay attacks impossible, even in edge cases such as identical reconstruction of a spent output.
- **Reward innovation.** Two-thirding halves nothing: every 210,000 blocks the reward becomes two-thirds of the previous reward, floored to whole units exactly as Bitcoin floors its halvings. The geometric series converges to the 31.5 million QUH cap over roughly 108 years, sustaining miner incentives through a smooth, predictable decay instead of a cliff.
- **Scalability and privacy.** A 2 MB block budget sized for post-quantum signature volume, five-minute confirmations, a bounded 2048-block difficulty controller, and one-time-address-friendly 512-bit Bech32m encoding balance throughput, verification cost, and unlinkability.

50 QUH

initial block reward

2/3

reward factor per epoch

**31.5 M
QUH**

total supply cap

8

decimal places (quphi)

This paper outlines the problem, the solution, and the potential applications of Qeuph QUH across industries such as finance, healthcare, and supply chain management. Sections 1 through 8 follow the protocol from cryptography to application; Appendix A records the shipped implementation parameters.

1. Introduction

Quantum computing presents an existential challenge to the cryptographic algorithms underlying most blockchain protocols. Estimates suggest that quantum computers capable of breaking widely used algorithms like secp256k1 and SHA2-256 may become viable soon. Existing blockchain systems lack the necessary safeguards to withstand such advancements. A fault-tolerant quantum computer running Shor's algorithm recovers ECDSA private keys from public keys in hours; harvested-and-decrypt archives of today's traffic become readable the moment such a machine exists. Blockchains are especially exposed because their public keys, signatures, and hashes are permanent, globally replicated evidence - a ledger cannot patch itself retroactively the way a website can rotate a certificate.

1.1 Gaps in Current Protocols

- Vulnerability to quantum attacks on digital signatures and hashing: virtually every production chain authenticates state changes with ECDSA or Schnorr over secp256k1/ed25519, and commits blocks with SHA-2 digests.
- Inefficiencies in transaction processing and scalability: long confirmation times, unbounded mempool behavior, and difficulty controllers that oscillate under hashrate swings.
- Limited mechanisms for sustaining long-term miner incentives: halving schedules drop rewards by 50% in a single block, producing sudden economic shocks and fee-market instability late in the emission curve.

1.2 The Qeuph QUH Solution

Qeuph QUH integrates three proven primitives into one coherent ledger. ML-DSA-87 is the NIST-standardized module-lattice signature (FIPS 204) at security category 5; SHA3-512 is the Keccak-family digest standardized in FIPS 202; and Bech32m (BIP-350) is the checksummed address encoding already battle-tested by Bitcoin taproot. The hybrid port of the QRL node architecture contributes the engineering scaffolding - chain management, peer-to-peer sync, wallet, and mining services

- rebuilt around these quantum-resistant primitives.

Component	Chosen primitive	Standard	Role
Digital signatures	ML-DSA-87	FIPS 204	Authorize every input
Hashing	Double SHA3-512	FIPS 202	Ids, Merkle, PoW, addresses
Address encoding	Bech32m, HRP quh	BIP-350	Checksummed addresses
Consensus	PoW (SHA3-512)	-	Nakamoto-style fork choice
Ledger	UTXO + txnonce	-	Balances and replay defense

Table 1: Qeuph QUH technology stack.

1.3 Roadmap of the Paper

Section 2 specifies transactions and the txnonce. Section 3 derives addresses. Section 4 covers blocks, proof of work, difficulty, and the two-thirding reward schedule. Section 5 analyzes security, Section 6 privacy, and Section 7 applications, before Section 8 concludes. Appendix A pins implementation parameters.

2. Transactions

A Qeuph QUH transaction is defined by its quantum-resistant design and structured validation process. The whitepaper-level model - inputs, outputs, a txnonce, lattice signatures, and double-hashed ids - is realized as a fixed-layout canonical byte encoding, which keeps parsing deterministic and removes any dependency on schema negotiation between nodes.

2.1 Key Features

- **Inputs and outputs.** Inputs are references to previously unspent outputs (a 64-byte transaction id plus a 32-bit index); outputs bind a value in quphi to a 64-byte address hash.
- **txnonce.** Ensures transaction uniqueness by incrementing for each new transaction of an address, preventing replay attacks independently of which UTXO is spent.
- **Digital signatures.** Each input carries an ML-DSA-87 signature over the double-SHA3-512 digest of the signature-less transaction prefix plus the input index, so every byte of the transaction - including the other inputs and their public keys - is committed by every signature.
- **Transaction ids.** Derived from double SHA3-512 hashing of the full serialization.

2.2 Transaction Flow

A payment moves through five stages, each independently checkable by any node without trusted setup:

- The user signs the transaction with their private key. A wallet seed expands deterministically to the ML-DSA-87 key pair, so the 32-byte seed is the only secret that must be protected.
- The transaction is broadcast to the network for validation.
- Nodes verify inputs, outputs, and the txnonce for integrity: the referenced UTXO must exist and be mature, the input's public key must hash to the UTXO owner, and the carried txnonce must equal the address's next expected value.
- The ML-DSA-87 signature is verified against the committed digest.
- Verified transactions are included in a block and hashed into the Merkle tree, whose root anchors them in the header.

Field	Encoding	Size	Notes
version	uint32 LE	4	1
in_count / out_count	uint32 LE	4 each	counts
prev_txid / prev_index	hash / uint32	64 + 4	UTXO reference
txnonce	uint64 LE	8	per-address counter
pubkey	raw ML-DSA-87	2592	owner of the UTXO
signature	raw ML-DSA-87	4627	over prefix + index
output value	uint64 LE	8	quphi (1 QUH = 10 ⁸)
output address hash	double SHA3-512	64	Bech32m-decodable

Table 2: Transaction field layout as implemented.

Signature economics. One ML-DSA-87 signature plus key is 7.2 KB - about 1,800 times a 72-byte ECDSA signature. Qeuph therefore budgets a 2 MB block (roughly 250 single-input transactions) on a five-minute cadence, and the reference node verifies a signature in roughly 20 μ s with the OpenSSL backend or 20 ms in pure Python.

3. Address Generation

Qeuph addresses compress a 2592-byte lattice public key into a human-typable, checksummed string while making the compression quantum-resistant. The pipeline is intentionally three steps so that each stage can be audited and replaced independently.

3.1 Process

- **Public key derivation.** Generated using ML-DSA-87, yielding a 2592-byte (5184 hexadecimal character) public key from a 32-byte seed.
- **Double hashing.** The public key is hashed twice using SHA3-512 for added security, producing a 512-bit digest.
- **Bech32m encoding.** The digest is encoded with the prefix quh, creating a 113-character address that detects up to four character entry errors.

3.2 Example

For a wallet seed, the derivation proceeds as follows: the ML-DSA-87 keygen expands the seed into the public key ab..... (2592 bytes); the double SHA3-512 hash 12..... compresses it to 64 bytes; and the Bech32m encoding yields the address quh11nt8... A test vector from the implementation:

Stage	Output	Length
Seed (secret)	hex seed	32 bytes
ML-DSA-87 public key	raw lattice key	2592 bytes
Double SHA3-512	address hash	64 bytes
Bech32m address	quh1<103 chars><6 checksum>	113 chars

Table 3: Address derivation sizes.

3.3 Importance of Bech32m

Bech32m improves usability and reduces errors during address entry. Unlike traditional formats, it incorporates a checksum for quick validation, ensuring secure and efficient transactions. Qeuph keeps the entire 512-bit hash as payload: address collisions then require breaking a full-width SHA3/Keccak preimage - a 2^{256} -bit quantum work factor - rather than merely colliding a 160-bit truncation. The cost is a longer string (113 characters), which wallets and QR codes absorb comfortably.

4. Blocks and Consensus Mechanism

Blocks follow the classic header-plus-transactions shape, with every field fixed-width and little-endian so that serialization is canonical. The consensus layer is deliberately conservative: proof of work over the header hash, bounded retargeting, and most-cumulative-work fork choice - the mechanism with the longest field-tested history in permissionless ledgers, now protected by a quantum-resistant hash.

4.1 Block Structure

Header field	Size	Meaning
version	4 B	protocol version
prev_hash	64 B	double SHA3-512 of parent header
merkle_root	64 B	pairwise double-SHA3-512 over txids
timestamp	8 B	seconds; monotonic, 2h future bound
bits	4 B	compact PoW target
height	8 B	distance from genesis
nonce	16 B	PoW search space

Table 4: The 168-byte block header. Block hash = double SHA3-512(header).

Block time is fixed at five minutes for faster confirmation times than ten-minute chains while keeping orphan rates low, and difficulty adjusts every 2048 blocks - approximately seven days. The first transaction of a block must be the coinbase, which embeds the block height, pays the miner, and becomes spendable only after 100 confirmations.

4.2 Consensus Algorithm

Qeuph QUH employs pure proof of work with SHA3-512: a block is valid when its hash, read as a 512-bit big-endian integer, is strictly less than the target encoded in the bits field. Retargeting multiplies the parent target by the ratio of measured to expected timespan, clamped to a factor of four in either direction, with all arithmetic in integers. Fork choice follows cumulative work. The hash function choice matters for the quantum era: Grover's algorithm at best halves effective hash security, and a 512-bit digest retains a comfortable margin even under that bound.

4.3 The Two-Thirding Mechanism

Block rewards start at 50 QUH and decrease by a factor of two-thirds every 210,000 blocks. Rewards are floored to whole quphi, exactly as Bitcoin floors its halvings, so issuance is always an exact integer computation:

```
reward(h) = floor( 50 QUH × (2/3)floor(h / 210000) ) yielding 50 → 33.33333333 → 22.22222222 → 14.81481481 → 9.87654320 → ... QUH
```

The schedule ensures sustained miner incentives over time and a predictable, gradual coin supply distribution. Because the decay is smooth rather than cliff-shaped, security-budget transitions are gentler than halving chains experience, and the emission horizon - about 108 years - is long enough for several generations of mining technology. The last non-zero reward, one quphi, is paid at height 11,130,000; thereafter miners earn fees only.

4.4 Total Supply: 31,500,000 QUH

The geometric series sums to at most three times the first epoch's issuance: $3 \times 50 \text{ QUH} \times 210,000 \text{ blocks} = 31,500,000 \text{ QUH}$, or 31.5 million QUH in words. Floor rounding leaves the exact total slightly below the cap - 31,499,999.8593 QUH - in the same way Bitcoin's shifted halvings never quite reach 21 million. The charts below show the epoch rewards and the cumulative emission converging on the cap.

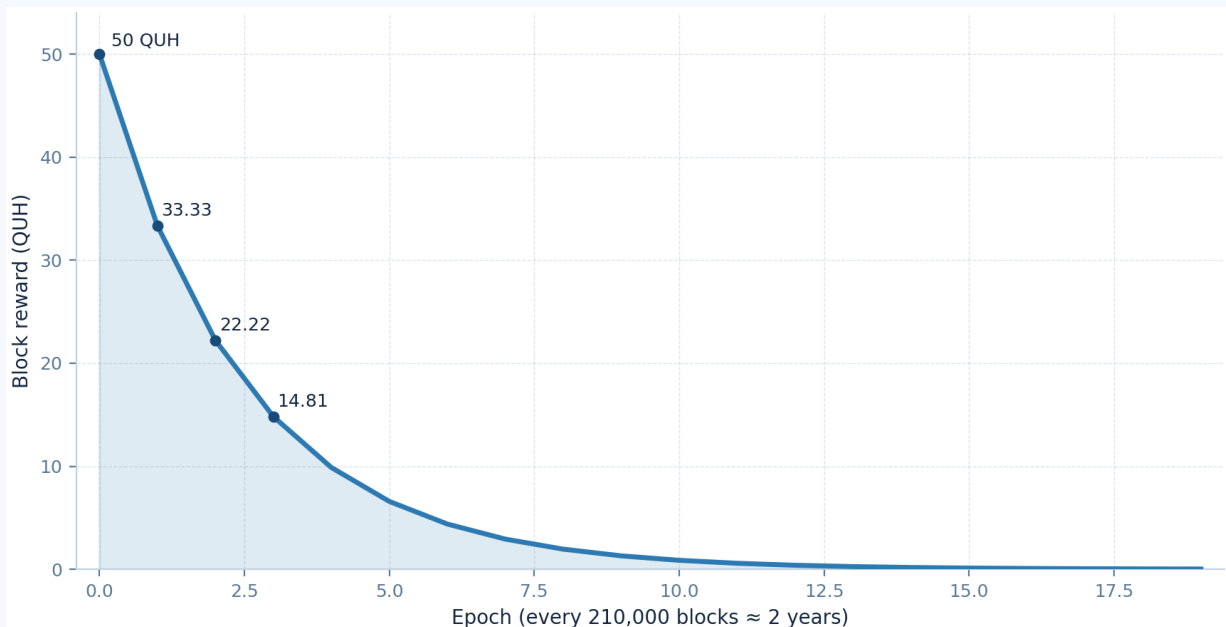


Figure 1: Block reward by epoch (first 20 of 54 epochs; one epoch = 210,000 blocks ≈ 2 years).

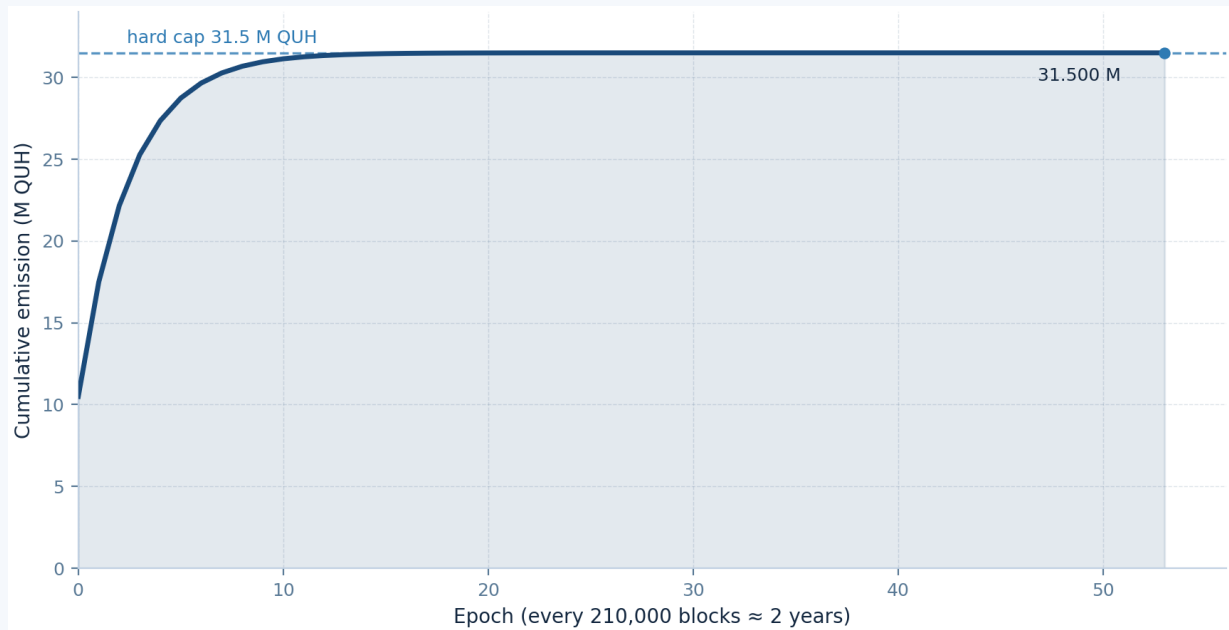


Figure 2: Cumulative emission approaching the 31.5 M QUH cap over 54 epochs (~108 years).

Epoch	Start height	Reward (QUH)	Cumulative (QUH)
0	0	50.00000000	10,500,000.000
1	210,000	33.33333333	17,499,999.999
2	420,000	22.22222222	22,166,666.666
3	630,000	14.81481481	25,277,777.776
4	840,000	9.87654320	27,351,851.848
5	1,050,000	6.58436213	28,734,567.895
10	2,100,000	0.86616069	30,963,969.705
20	4,200,000	0.00014994	31,499,999.856
53	11,130,000	0.00000001	31,499,999.859
54	11,340,000	0 (fees only)	31,499,999.859

Table 5: Two-thirding emission schedule (exact values from the implementation).

5. Security Considerations

Security in Qeuph is layered: the signature layer resists quantum forgery, the transaction layer resists replay and malleation, the consensus layer resists reorganization and timestamp games, and the address layer resists collision attacks. This section evaluates the signature layer against alternatives and walks the mitigation of each named attack class.

5.1 Comparison of ML-DSA-87

Feature	ML-DSA-87	Other lattice algorithms
Quantum resistance	High (category 5)	Variable
Efficiency	Optimized (fast verify)	Mixed
Adoption	FIPS compliant	Limited

Table 6: ML-DSA-87 versus other lattice-based candidates.

ML-DSA-87 is parameterized for NIST security category 5 - at least as hard to break as AES-256 - and its module-lattice assumptions (M-LWE and M-SIS) have withstood two decades of cryptanalytic effort. Because it is a NIST standard with an ACVP validation program, implementations can be certified, which no earlier lattice candidate could offer.

5.2 ML-DSA-87 Parameters

Parameter	Value	Meaning
q	8380417	modulus $2^{23} - 2^{13} + 1$
(k, l)	(8, 7)	matrix A dimensions
eta	2	secret coefficient range
tau	60	challenge weight
gamma1	2^{19}	mask coefficient range
gamma2	(q-1)/32	rounding range
public key	2592 bytes	rho + t1
signature	4627 bytes	c-tilde + z + hint
security	category 5	$\geq$ AES-256

Table 7: FIPS 204 ML-DSA-87 parameter set as implemented.

5.3 Attack Mitigations

- **Side-channel attacks.** Addressed through secure hardware and implementation practices; the reference implementation defaults to the hedged signing variant of FIPS 204, which randomizes the nonce derivation so repeated signing of the same message leaks nothing.
- **Replay attacks.** Mitigated by the txnonce: a transaction carrying a stale per-address counter is rejected outright, and the counter is part of the signed digest, so it cannot be edited without invalidating the signature.
- **Quantum threats.** Neutralized by ML-DSA-87's lattice-based hardness: forging requires solving module-lattice problems whose best known quantum algorithms offer only polynomial speedups, not the exponential break Shor's algorithm delivers against elliptic curves.

5.4 A Forging Attempt, Concretely

Consider a quantum computer attempting to forge a transaction. ML-DSA-87's lattice-based hardness ensures the computation would take time exponential in the lattice dimension, rendering the attack infeasible. Breaking one signature means finding a short vector in a 2048-dimensional module lattice over $q = 8,380,417$; the best known quantum algorithms for this problem (BKZ with Grover-accelerated subroutines) still require on the order of 2^{140} operations at category 5 parameters - against secp256k1, the same machine needs only hours.

6. Privacy

Qeuph QUH is pseudonymous by design: the ledger records commitments to hashes, never identities. Because addresses are 512-bit hashes of lattice public keys, and keys are revealed only when an input is spent, the default posture hides public keys until first spend - the same house-rule Bitcoin learned the hard way, enforced here by cheap, disposable addresses.

6.1 Features

- **Anonymous public keys.** Transactions are unlinkable to real-world identities; on-chain data binds only address hashes to values, and the 512-bit hash space makes brute-force linkage or collision attacks impractical even for state-sized adversaries.
- **Key pair rotation.** A new key pair is recommended for each transaction to enhance anonymity. Seed-derived indexes make rotation a one-line wallet operation, and change addresses are automatic: every spend returns the surplus to a fresh output rather than the paying address.

- **No metadata leakage.** The canonical transaction format carries no sender hints, memo fields, or client fingerprints; the txnonce is the only per-address state and it is order information, not identity information.

Two caveats keep the model honest. First, coinbase outputs and long-lived merchant addresses still cluster value by reuse, so rotation discipline matters as much as the cryptography. Second, network-level analytics - IP correlation across the P2P layer - is out of scope for the ledger itself and belongs to transport anonymization. Neither weakness is unique to Qeuph, and both are orthogonal to its quantum-resistance goal: privacy here means no identity binding in the protocol, not full anonymity.

7. Real-World Use Cases

Three sectors gain the most from a ledger whose signatures and hashes survive quantum adversaries, because their records must stay verifiable for decades - longer than any comfortable estimate of when large quantum computers arrive.

7.1 Finance

Qeuph QUH ensures secure and fast cross-border payments resistant to quantum threats. Settlement records written today must remain provable when banks reconcile them in the 2050s; a five-minute cadence, integer difficulty control, and 31.5 million unit hard cap make the asset usable both as a settlement rail and as a long-horizon store of value. The txnonce also gives payment processors strict per-account ordering - sequence numbers they already understand from correspondent banking.

7.2 Healthcare

Medical records carry statutory retention duties of twenty-five years or more. Anchoring record hashes on Qeuph protects them from unauthorized access and later tampering: an audit entry written under ML-DSA-87 remains verifiable without any migration event, and the 512-bit address hash prevents patient-identifier collisions across institutional namespaces.

7.3 Supply Chain

Provides immutable and secure tracking of goods, ensuring data integrity across global networks. Provenance attested at manufacture time stays provable through the whole custody chain, counterfeit lot recalls can be pinned to the exact hand-off where custody diverged, and multi-input transactions let counterparties co-sign a custody transfer in a single on-chain object.

8. Conclusion

Qeuph QUH addresses the pressing need for quantum-resistant blockchain solutions. By integrating cutting-edge cryptography - ML-DSA-87 per FIPS 204 and double SHA3-512 - with proven mechanisms such as two-thirding rewards, UTXO-plus-txnnonce validation, and Bech32m addressing, it ensures long-term security, efficiency, and scalability. The protocol ships as a complete, self-contained node suite whose pure-Python FIPS 204 implementation is cross-verified against OpenSSL's standardized ML-DSA, giving auditors two independent paths to every signature on the chain. Join us in building a future-proof decentralized ecosystem.

Appendix A: Implementation Specification

The reference implementation (Python 3.10+, no mandatory dependencies) pins the following consensus parameters. Regtest and testnet profiles vary only in ports, address prefixes, and genesis configuration.

Parameter	Value	Parameter	Value
Block time	300 s	Retarget interval	2048 blocks
Retarget clamp	$\times/\div 4$	Max block size	2,000,000 B
Reward interval	210,000 blocks	Initial reward	50 QUH
Reward rule	$\text{floor}(r \times 2/3)$	Supply cap	31,500,000 QUH
Exact emission	31,499,999.8593 QUH	Decimals	8 (quphi)
Coinbase maturity	100 blocks	Future time bound	2 hours
P2P port (mainnet)	19090	JSON-RPC port	19091
Network magic	QUH! (0x51554821)	Address HRP	quh
Genesis timestamp	1790812800 (2026-10-01 UTC)	Genesis bits	0x3D0FFFFFF

Table 8: Shipped consensus and network parameters.

Mainnet genesis. The shipped genesis block is pinned with its winning proof-of-work nonce 355026620, mined at the initial difficulty (bits 0x3D0FFFFFF). Its double SHA3-512 id is 0000000d2f105b239cd085e9d4bd7fa087dc6a085ee37b3842d539ab9c974247fa8c4000695807b0a5d306d40c713e32b130387f72b64a00bf2b0c9953b179b1; every node reconstructs this exact block from the pinned constant, so chain identity is fixed at launch.

Transaction and block encodings are exactly the layouts of Tables 2 and 4. The wire protocol frames commands with a magic header, a double-SHA3-512 payload checksum, and JSON payloads; initial block download is headers-first; the wallet keystore seals the 32-byte master seed with AES-256-GCM under PBKDF2-HMAC-SHA3-512 with 60,000 iterations. The full test suite - including FIPS 204 cross-verification against OpenSSL, txnonce replay and ordering rules, difficulty clamping, reorganisation, and an end-to-end daemon integration test - ships with the source.

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